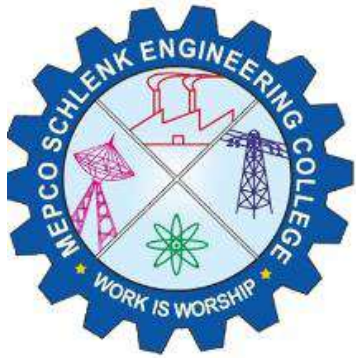




IEEE STB-29861 EPICS IN IEEE

<https://epics.ieee.org/>



Engineering Projects in Community Service

Engineering Solutions to real life Community Challenges

- **Define the problem**
- **Design solutions and optimize**
- **Build and implement**
- **Testing**
- **Deploying**

Students Benefits

Professional skills such as

- Interdisciplinary learning approach integrating STEM for real world problems
- Communication,
- Collaboration with professionals, NGOs, local community
- Leadership
- Project management
- Time management

Benefits for Students

- **Funding:** Financial support ranging from **\$500 to \$10,000 USD**.
- **Global Networking:** Connection with IEEE professionals and international peers.
- **Recognition:** Opportunity to present results at [IEEE conferences via travel grants](#).

Service Learning

- An **opportunity** for **university students** to work **collaboratively** with **engineering professionals** and **non-profit organizations** to develop **innovative solutions** that **transform communities** worldwide. The goal is to encourage learning not only in the community but **with the community**.
- **The EPICS in IEEE proposal process builds in pedagogical best practices so that only those proposals with strong service-learning pedagogy are accepted for funding.**

EPICS in IEEE uses a structured rubric to evaluate project proposals based on **community partnership, engineering design, and project feasibility.**

Key criteria include a **clear problem statement, strong nonprofit partnership, detailed technical solutions, appropriate budget/timeline, and measurable student learning outcomes.**

- **Mission:** Empowers students to work with technical professionals for community benefit.
- **Approach:** Uses a human-centered design process to solve real-world problems.
- **IEEE Support:** Provides **seed funding, mentorship, and global visibility.**

Human Centred Design (HCD) Phases

- **Identification:** Define the specific need with stakeholders
- **Requirements:** Specify exactly what the user and context need
- **Ideation:** Generate multiple potential solutions before picking the best one.
- **Prototyping:** Build low-cost versions for early feedback and testing.
- **Refinement:** Iterative cycles where the design is modified based on user input

Eligibility Criteria

- **Team Composition:** Must include students.
- **IEEE Membership:** At least **one team member** must be an active IEEE member.
- **Lead Role:** Can be a **student member, faculty advisor, or professional member.**
- **Institutional Support:** Encouraged to work with **local IEEE Student Branches or Sections.**
- **Volunteer Engagement:** Teams often involve **non-student IEEE volunteers for technical mentorship.**

The Community Partner (NPO/NGO)

- **Requirement:** A project **must** have a non-profit organization or NGO as a partner.
- **Partner's Role:** Connects the team with end-users and defines the local need.
- **Sustainability:** The partner must agree to help maintain the solution after deployment.
- **Letters of Support:** A support letter from the NGO is highly recommended for the application.
- **Pre-Survey:** Partners are often asked to gauge current status before the project starts.

Scope of Technology

- **Technical Fit:** Solutions must fall within the [IEEE fields of interest](#) (Electrical, Electronics, etc.).
- **Innovation:** Projects should apply technology in new ways to solve specific societal challenges.
- **Example Areas:** IoT, AI, autonomous systems, and renewable energy.

Environment & Sustainability

- **Target:** Local environmental challenges and climate change mitigation.
- **Goals:** Clean water, renewable energy, and waste management.
- **Examples:** **Solar-powered irrigation or microplastic detection** systems.
- **Innovation:** Using low-cost sensors to monitor environmental parameters.
- **Commitment:** Projects must show a **direct benefit to the local ecology or resources.**

Human Services

- **Target:** Basic human needs in underserved communities.
- **Goals:** Improving health services, food security, and basic infrastructure.
- **Examples:** Modular clinics, vaccine refrigeration, or sanitation systems.
- **Social Good:** Focuses on "engineering social innovation" through partnerships.
- **Scale:** Often addresses challenges affecting large populations in rural areas.

Drafting the Proposal

- **Project Title:** Must be concise (under 100 characters).
- **Problem Statement:** Describe the community gap (limit 1,000 characters).
- **Solution Description:** Detail the technical approach (limit 10,000 characters).
- **Inputs:** Describe necessary personnel, materials, and software.
- **Outputs:** List deliverables that will be created and deployed.

Budgeting Guidelines

- **Allowable Costs:** Prototyping materials, project supplies, and local travel.
- **Strict Prohibitions:** Honorariums, stipends, or personal computers (laptops).
- **Capital Equipment:** General office equipment or construction costs are ineligible.
- **Justification:** Every item must have a clear explanation of its need for the project.
- **Tools:** Use the official [EPICS Budget Template](#) for submission.

Project Timeline & Duration

- **Maximum Length:** Projects must be completed within **12 months**.
- **Phasing:** Large projects should be broken into one year phases with clear yearly deliverables.
- **Milestones:** Must include key gates for design, testing, and delivery.
- **Deadline:** Proposals often have specific cycles (e.g., May 1st for summer cycles).
- **Efficiency:** Use a "time-boxed" approach for prototypes to save resource

Core Evaluation Components (Project Proposal Rubric)

The EPICS in IEEE Committee evaluates proposals based on the following key areas:

- Problem Summary and Justification:** A clear description of the community problem, the target end-users, and why the project is necessary.
- Community Partnership:** Involvement of a specific nonprofit organization or NGO (community partner) that is actively involved in the project, not just a beneficiary.
- Engineering Design Process:** Application of engineering principles to design, build, and deploy the solution, falling within the scope of IEEE.
- Proposed Technological Solution:** Feasibility, innovation, and technical quality of the proposed technology.
- Project Planning and Budget:** A realistic timeline and a detailed, justified budget that complies with funding regulations.
- Team Composition:** A team with the necessary skills to complete the project, including at least one IEEE member.
- Impact and Assessment Plan:** Defined metrics to assess both the impact on the community and the learning outcomes for the students

Post-Approval Reporting

- **Quarterly Reports:** Mandatory technical and financial updates every 3 months.
- **Final Report:** Must include a financial summary and final impact data.
- **Photo Releases:** Teams must facilitate photo-release and liability waivers.
- **Publicity:** Partners should assist in publicizing project success.
- **Fund Transfer:** Funds are typically managed through the local IEEE Section.

Registration Checklist

- Active IEEE Membership for at least one team member.
- Signed agreement or support letter from an NGO/NPO partner.
- Detailed itemized budget using the official template.
- Technical design diagrams or flowcharts ready for upload.
- Completed proposal on the [IEEE Grant Management Platform](#).
- <https://epics.ieee.org/resources/> **Course on IEEE EPICS**

Useful Links & Sanctioned Projects

- **Official Site:** epics.ieee.org
- **Apply Here:** IEEE EPICS Application Portal
- **Training:** [Proposal Writing & Budgeting Course](#).
- **Example 1:** Smart Braille System (Pakistan) — Funded \$9,155 for visually impaired education.
- **Example 2:** Solar-Powered Water Pumps (Uganda) — Environmental sustainability in rural areas

Useful Links & Sanctioned Projects

- **Budget Template:** [Download Training/Template](#)

Past Sanctioned Projects:

- **Solar-Powered Water Pumps (Uganda):** Sustainable water conservation in remote villages.
- **Smart Outdoor Wheelchair (India):** IoT-based mobility aid for the paralyzed.
- **Solar Refrigeration for Vaccines (India):** Modular systems for rural health facilities.
- **Microplastic Detection System (Pakistan):** Innovative sensing to address water pollution.

Thank You